

Market Risk Stress Testing

Historical and Hypothetical Stress Analysis of a
\$100 Million Multi-Asset Portfolio

Portfolio Stress Testing | P&L Attribution | Cross-Scenario Risk Analysis

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Executive Summary

This report evaluates the stress resilience of a hypothetical \$100 million multi-asset portfolio using complementary historical and hypothetical stress-testing frameworks. The portfolio contains exposures across Equity, Rates, Credit, FX, and Commodity asset classes. Historical stress testing replays realized instrument returns from selected periods of market disruption, while hypothetical stress testing applies predefined shocks to modeled risk factors and translates those shocks into position- and portfolio-level P&L.

The analysis evaluates seven stress scenarios: three historical episodes and four hypothetical scenarios. Portfolio outcomes range from a gain of 4.47% of NAV under Growth / Risk-On Rally to a loss of 23.09% of NAV during the 2022 Inflation / Rates Sell-Off. Among the hypothetical scenarios, Stagflation Shock produces the largest loss at 15.00% of NAV, followed by Rates Higher for Longer at 12.95% of NAV. The COVID-19 Market Crash generates a historical loss of 16.39% of NAV, while the 2023 Regional Banking Stress produces a gain of 2.45% of NAV.

The results identify Equity concentration as the portfolio's largest structural source of downside, reflecting its 50% allocation to the asset class. However, Equity weakness alone does not determine overall portfolio stress severity. The most damaging outcomes occur when losses extend across multiple major exposures. During the 2022 Inflation / Rates Sell-Off and the hypothetical Stagflation Shock, Equity losses coincide with adverse fixed-income performance, causing exposures that might otherwise provide diversification to reinforce portfolio losses.

A key finding is that the diversification benefit of Rates is strongly regime-dependent. Treasury exposure provides meaningful loss mitigation when falling yields accompany risk-asset weakness, as observed during the COVID-19 Market Crash and modeled under Global Risk-Off. In inflationary or rising-rate environments, however, the same duration exposure becomes a source of loss. Credit can further amplify these outcomes through the combined effects of underlying rate movements and changes in credit spreads, while gold provides a smaller partial offset in selected scenarios.

Taken together, the historical and hypothetical frameworks show that the portfolio is most vulnerable not simply to large individual market shocks, but to adverse alignment across its underlying economic risk drivers. The central risk-management implication is that diversification should be evaluated not only by the number of instruments or asset classes held, but also by how those exposures are expected to interact across different market regimes.

1. Portfolio Composition and Risk Exposure

1.1 Portfolio Overview

The stress-testing framework is applied to a hypothetical multi-asset portfolio with a total NAV of \$100 million. The portfolio contains ten positions spanning five broad asset classes: Equity, Rates, Credit, Commodity, and Foreign Exchange (FX).

The portfolio contains \$50 million of Equity exposure through U.S. and European equity ETFs, \$25 million of Rates exposure through intermediate- and long-duration U.S. Treasury ETFs, \$15 million of Credit exposure through investment-grade and high-yield corporate bond ETFs, \$5 million of gold exposure, and \$5 million of foreign-exchange exposure.

Table 1. Portfolio Positions

Instrument	Asset Class	Exposure (USD)	% NAV
SPY	Equity	\$25,000,000	25.0%
QQQ	Equity	\$15,000,000	15.0%
VGK	Equity	\$10,000,000	10.0%
IEF	Rates	\$15,000,000	15.0%
TLT	Rates	\$10,000,000	10.0%
LQD	Credit	\$10,000,000	10.0%
HYG	Credit	\$5,000,000	5.0%
GLD	Commodity	\$5,000,000	5.0%
EURUSD	FX	\$3,000,000	3.0%
USDJPY	FX	\$2,000,000	2.0%
Total		\$100,000,000	100.0%

Equity represents the largest allocation at 50% of NAV, followed by Rates at 25%, Credit at 15%, Commodity at 5%, and FX at 5%. Portfolio outcomes therefore depend not only on the magnitude of shocks to individual markets, but also on how gains and losses across these asset classes interact under different regimes.

1.2 Asset-Class Allocation

- Equity - 50% of NAV: SPY (25%), QQQ (15%), and VGK (10%) form the portfolio's principal source of directional growth-market exposure.
- Rates - 25% of NAV: IEF (15%) and TLT (10%) provide intermediate- and long-duration U.S. Treasury exposure.
- Credit - 15% of NAV: LQD (10%) and HYG (5%) introduce both underlying interest-rate and corporate-spread sensitivity.
- Commodity - 5% of NAV: GLD represents gold exposure and a potential diversification source.
- FX - 5% of NAV: EURUSD (3%) and USDJPY (2%) introduce direct currency sensitivity.

This allocation creates an important distinction between portfolio weight and portfolio risk. For fixed-income instruments in particular, duration and spread sensitivity influence stress P&L in addition to the size of the underlying exposure.

1.3 Risk-Factor Mapping

The hypothetical framework maps each position to one or more market risk factors. Three broad types are modeled: direct-return factors, interest-rate factors transmitted through modified duration, and credit-spread factors transmitted through spread duration.

Table 2. Instrument-to-Risk-Factor Mapping

Instrument	Asset Class	Risk Factor(s)	Type
SPY	Equity	SPX	Direct return
QQQ	Equity	NASDAQ	Direct return
VGK	Equity	EU_EQUITY, EURUSD	Direct return
IEF	Rates	USD_7Y_RATE	Interest rate
TLT	Rates	USD_20Y_RATE	Interest rate
LQD	Credit	USD_10Y_RATE, IG_SPREAD	Rate + spread
HYG	Credit	USD_5Y_RATE, HY_SPREAD	Rate + spread
GLD	Commodity	GOLD	Direct return
EURUSD	FX	EURUSD	Direct return
USDJPY	FX	USDJPY	Direct return

Scenario Shock -> Risk Factor -> Position P&L -> Asset-Class P&L -> Portfolio P&L

Primary factor sensitivities are generally 1.0. VGK additionally carries a 0.35 sensitivity to EURUSD. Credit positions use a two-factor representation so rate and spread contributions can be distinguished.

1.4 Fixed-Income Sensitivity Assumptions

Modified Duration

Modified duration approximates the percentage price change of a fixed-income instrument for a change in its underlying yield.

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y$$

Spread Duration

Spread duration approximates the price response of corporate credit instruments to changes in credit spreads.

$$\frac{\Delta P}{P} \approx -SD \Delta s$$

Table 3. Fixed-Income Sensitivity Assumptions

Instrument	Asset Class	Risk Factor	Mod. Duration	Spread Factor	Spread Duration
IEF	Rates	USD_7Y_RATE	7.3	-	-
TLT	Rates	USD_20Y_RATE	15.8	-	-
LQD	Credit	USD_10Y_RATE	8.1	IG_SPREAD	7.6
HYG	Credit	USD_5Y_RATE	3.5	HY_SPREAD	3.3

Duration parameters are static model inputs. The approach is first-order and does not incorporate convexity or other nonlinear effects; these limitations are discussed in Section 6.

2. Stress Testing Methodology

2.1 Stress Testing Framework

The framework evaluates the portfolio under two complementary approaches: hypothetical stress testing and historical stress testing. Both estimate the impact of stressed market conditions on current exposures, but they differ in how market movements are defined and transmitted to positions.

Hypothetical stress begins with predefined shocks to underlying risk factors. Each shock is translated into position-level P&L using assigned sensitivity and, where applicable, modified duration or spread duration.

Historical stress instead measures actual instrument returns over selected historical windows and applies those observed returns to the portfolio's current exposures.

Hypothetical: Scenario -> Risk-Factor Shocks -> Position Sensitivities -> Position P&L -> Portfolio P&L

Historical: Stress Window -> Observed Instrument Returns -> Current Exposures -> Position P&L -> Portfolio P&L

Neither approach is a forecast. Each is a conditional analysis of the current portfolio under a specified stressed environment.

2.2 Hypothetical Stress Methodology

For position i and factor f , stressed P&L depends on exposure, scenario shock, assigned factor sensitivity and, for fixed-income positions, the relevant duration measure.

$$\text{Direct return: } \text{PnL}_{i,f} = E_i \beta_{i,f} \Delta F_f$$

$$\text{Rate risk: } \text{PnL}_{i,f} \approx -E_i D_{i,f} \beta_{i,f} \Delta y_f$$

$$\text{Spread risk: } \text{PnL}_{i,f} \approx -E_i S D_{i,f} \beta_{i,f} \Delta s_f$$

All percentage shocks are represented in decimal form; a 100-basis-point change is 0.01. A positive yield or spread shock produces a negative first-order price effect.

2.2.1 Direct-Return Risk Factors

Direct-return shocks are used for SPX, NASDAQ, EU_EQUITY, GOLD, EURUSD and USDJPY. For example, a \$25 million SPY position with unit sensitivity under a -20% SPX shock generates a -\$5.0 million stress P&L.

2.2.2 Interest-Rate Risk Factors

For a \$10 million TLT position with modified duration 15.8, a +100 bps 20-year yield shock produces an estimated -\$1.58 million first-order loss.

2.2.3 Credit-Spread Risk Factors

LQD and HYG combine rate and spread components, allowing the framework to distinguish losses caused by movements in the Treasury curve from losses caused by changes in credit risk premia.

2.2.4 Position and Portfolio Aggregation

$$\begin{aligned} \text{PnL}_i &= \sum_f \text{PnL}_{i,f} \\ \text{PnL}_{\text{portfolio}} &= \sum_i \text{PnL}_i \\ \text{PnL}_{\text{NAV}} &= \frac{\text{PnL}_{\text{portfolio}}}{\text{NAV}} \times 100 \end{aligned}$$

With a \$100 million NAV, a \$15 million loss corresponds to -15.0% of NAV. The same normalization is used for position-, asset-class-, and risk-factor-level contributions.

2.2.5 Hypothetical P&L Attribution

Risk-Factor P&L -> Position P&L -> Asset-Class P&L -> Portfolio P&L

The decomposition is additive. Subject to rounding, aggregation by risk factor, position, or asset class recovers the same portfolio-level result. Sensitivities are deterministic model inputs rather than statistically calibrated betas.

2.3 Historical Stress Methodology

Historical stress testing evaluates how the current portfolio would have performed under market movements observed during selected historical periods. Realized instrument returns are applied directly to current USD exposures.

2.3.1 Historical Scenario Definition

The historical scenarios are the COVID-19 Market Crash, the 2022 Inflation / Rates Sell-Off, and the 2023 Regional Banking Stress. Current portfolio exposures are held constant across all three.

2.3.2 Historical Return and P&L Calculation

$$R_{i,h} = \frac{p_{i,h}^{\text{end}}}{p_{i,h}^{\text{start}}} - 1$$
$$\text{PnL}_{i,h} = E_i R_{i,h}$$

A \$25 million position experiencing a -30% historical return generates a -\$7.5 million replay P&L.

2.3.3 Portfolio Aggregation and Attribution

$$\text{PnL}_{\text{portfolio},h} = \sum_i \text{PnL}_{i,h}$$

Historical attribution is reported at position and asset-class levels. Risk-factor attribution is not applied because the framework directly uses realized instrument returns.

2.3.4 Interpretation of Historical Stress

Historical replay preserves cross-asset relationships that actually occurred, but results depend on the selected window and should be interpreted as empirical reference scenarios rather than forecasts or a complete future loss distribution.

2.4 Scenario Design and Selection

The framework contains seven scenarios: four hypothetical scenarios and three historical scenarios. They are designed to expose the portfolio to materially different risk channels rather than variations of a single adverse environment.

Table 4. Hypothetical Scenario Set

Scenario	Primary Stress Theme	Key Risk Channels	Intended Portfolio Test
Global Risk-Off	Broad deterioration in risk appetite	Equity declines; spread widening; safe-haven rates/gold; FX	Conventional flight-to-quality
Stagflation Shock	Weak growth + persistent inflation	Equity losses; higher rates; wider spreads; commodity/FX	Growth and duration pressure together
Rates Higher for Longer	Persistent upward pressure on rates	Treasury yield increases; duration losses; credit repricing	Sensitivity to restrictive monetary conditions
Growth / Risk-On Rally	Improving growth and risk appetite	Equity gains; higher yields; spread tightening	Upside participation and defensive drag

Table 5. Historical Scenario Set

Scenario	Stress Window	Primary Market Theme	Intended Portfolio Test
COVID-19 Market Crash	19 Feb 2020 - 23 Mar 2020	Rapid global risk-asset sell-off and flight to safety	Extreme growth/liquidity shock
2022 Inflation / Rates Sell-Off	3 Jan 2022 - 14 Oct 2022	Inflation, tightening, simultaneous equity/bond weakness	Breakdown of equity-rate diversification
2023 Regional Banking Stress	8 Mar 2023 - 24 Mar 2023	Banking stress, falling rates, defensive demand	Financial-sector shock with offsets

Conventional Risk-Off: Equity Losses + Rate Gains

Inflationary / Stagflationary Stress: Equity Losses + Rate Losses

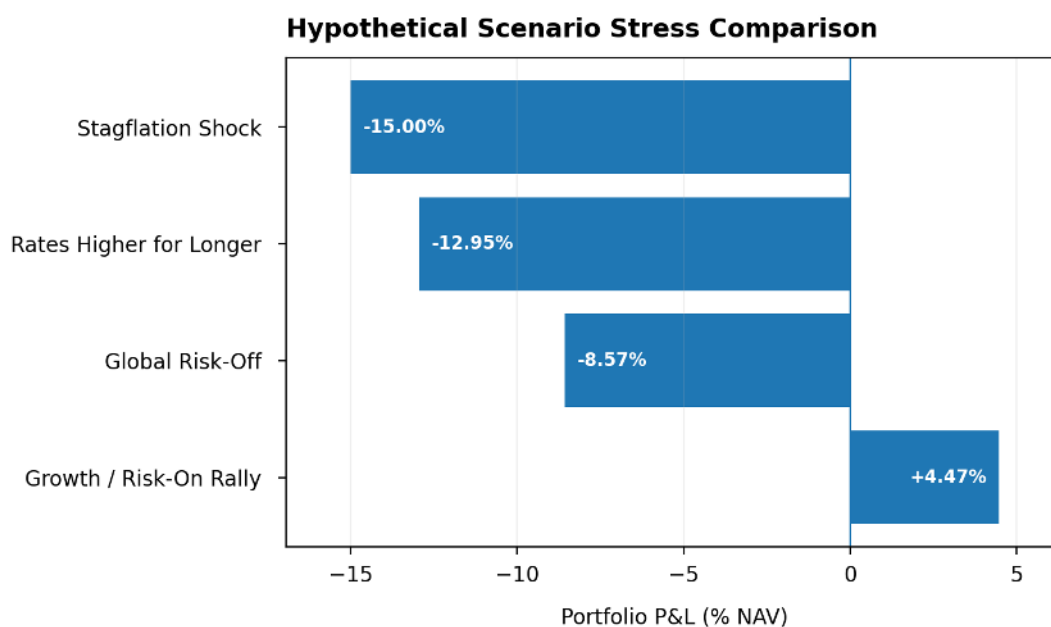
3. Hypothetical Stress Testing Results

3.1 Portfolio-Level Scenario Results

Table 6. Hypothetical Stress Scenario Results

Scenario	Portfolio Stress P&L	P&L (% NAV)
Stagflation Shock	-\$15.00 million	-15.00%
Rates Higher for Longer	-\$12.95 million	-12.95%
Global Risk-Off	-\$8.57 million	-8.57%
Growth / Risk-On Rally	+\$4.47 million	+4.47%

Figure 1. Hypothetical Scenario Stress Comparison



Stagflation Shock is the most severe hypothetical scenario, producing a \$15.0 million loss, or 15.00% of NAV. Simultaneous Equity weakness, rising rates, and wider credit spreads create losses across several major exposures; gold provides only a partial offset.

Rates Higher for Longer produces the second-largest hypothetical loss at 12.95% of NAV. Duration-related losses become particularly important as rising yields reduce the value of Treasury and investment-grade credit positions.

Global Risk-Off produces an 8.57% loss. Severe Equity losses are partially offset by gains in Treasury positions and gold, illustrating a more conventional flight-to-quality regime.

Growth / Risk-On Rally generates a 4.47% gain. Strong Equity performance more than offsets losses from duration, Credit, and gold.

3.2 Cross-Scenario Asset-Class Attribution

Table 7. Hypothetical Asset-Class Attribution (% NAV)

Asset Class	Global Risk Off	Stagflation	Rates Higher	Risk-On Rally
Equity	-11.23%	-10.27%	-5.84%	+8.12%

Rates	+2.51%	-3.11%	-4.80%	-2.85%
Credit	-0.23%	-2.14%	-1.99%	-0.63%
Commodity	+0.50%	+0.60%	-0.30%	-0.25%
FX	-0.12%	-0.07%	-0.02%	+0.07%
Portfolio	-8.57%	-15.00%	-12.95%	+4.47%

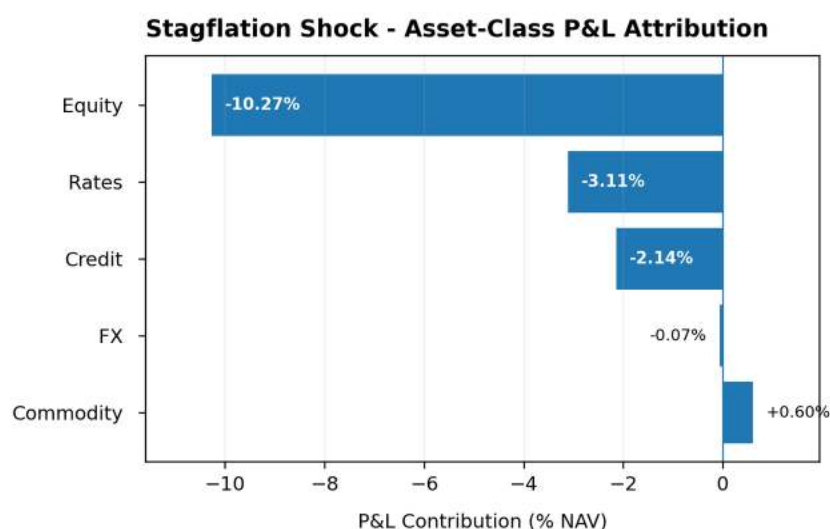
Equity is the dominant source of directional portfolio risk across the hypothetical scenarios. The behavior of Rates is strongly regime-dependent: +2.51% of NAV under Global Risk-Off, but -3.11% under Stagflation and -4.80% under Rates Higher for Longer.

Global Risk-Off: Equity down + Rates up | Stagflation / Rates Stress: Equity down + Rates down

Credit contributes negatively in all four hypothetical scenarios. Gold provides positive diversification under Global Risk-Off and Stagflation but remains small relative to larger Equity, Rates, and Credit exposures. FX has the smallest direct portfolio impact.

3.3 Stagflation Shock: Asset-Class Decomposition

Figure 2. Stagflation Shock - Asset-Class P&L Attribution



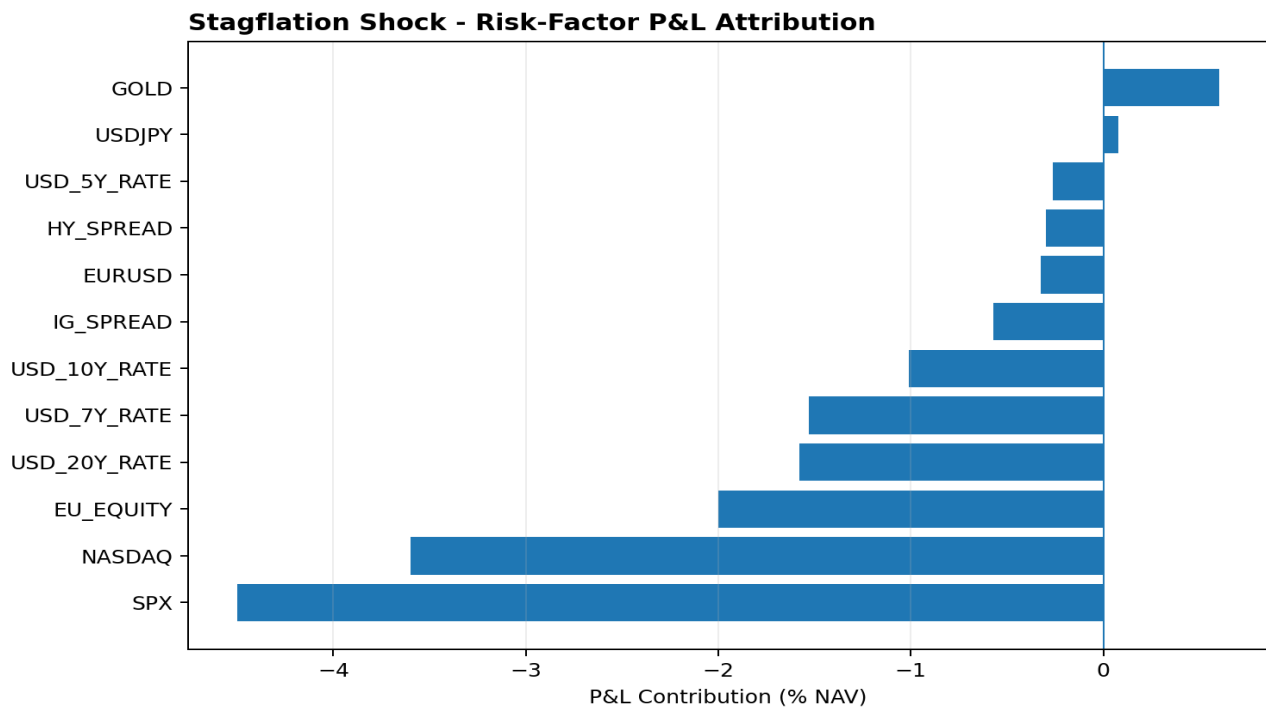
Equity (-10.27%) + Rates (-3.11%) + Credit (-2.14%) + FX (-0.07%) + Commodity (+0.60%) = Portfolio (-15.00%)

Equity contributes approximately -\$10.28 million and is the dominant loss source. Rates add -\$3.11 million and Credit -\$2.14 million. FX is small at -\$70,000. Commodity is the only meaningful positive asset class, with GLD contributing +\$600,000.

The scenario therefore illustrates cross-asset loss concentration: the portfolio's principal defensive fixed-income allocation becomes aligned with Equity losses, while the smaller gold allocation cannot materially change the overall outcome.

3.4 Stagflation Shock: Risk-Factor Decomposition

Figure 3. Stagflation Shock - Risk-Factor P&L Attribution



SPX contributes -\$4.50 million, NASDAQ -\$3.60 million, and EU_EQUITY -\$2.00 million. Together, these three Equity factors account for -\$10.10 million.

Rate factors are the next major source of loss: USD_20Y_RATE contributes -\$1.58 million through TLT and USD_7Y_RATE contributes -\$1.53 million through IEF. LQD and HYG add both rate and spread losses. EURUSD contributes approximately -\$325,000, USDJPY +\$80,000, and GOLD +\$600,000.

Equity-Factor Losses + Rate-Factor Losses + Spread Losses + FX Effects + Gold Offset = Portfolio Stress P&L

The decomposition distinguishes portfolio allocation from underlying risk concentration: diversification by instrument count does not necessarily imply diversification by economic risk driver.

3.5 Hypothetical Stress Findings

1. Equity is the dominant source of directional risk, reflecting the portfolio's 50% allocation.
2. The diversification benefit of Rates is strongly scenario-dependent.
3. Portfolio stress becomes most severe when multiple risk channels become adversely aligned.

Equity Weakness + Rising Rates + Credit Deterioration -> Diversification Breakdown

4. Historical Stress Testing Results

Historical stress testing complements the hypothetical analysis by applying realized market movements from selected periods of financial stress to the portfolio's current exposures.

4.1 Historical Scenario Comparison

Table 8. Historical Stress Scenario Comparison

Historical Scenario	Stress Window	Portfolio P&L	P&L as % NAV
2022 Inflation / Rates Sell-Off	2022-01-03 to 2022-10-14	-\$23.09m	-23.09%
COVID-19 Market Crash	2020-02-19 to 2020-03-23	-\$16.39m	-16.39%
2023 Regional Banking Stress	2023-03-08 to 2023-03-24	+\$2.45m	+2.45%

The 2022 Inflation / Rates Sell-Off is the most severe historical scenario. The ranking demonstrates that the most damaging episode is not necessarily the one with the sharpest Equity collapse; losses become more severe when Equity and fixed-income exposures decline simultaneously.

4.2 COVID-19 Market Crash

The COVID-19 Market Crash produces a portfolio loss of approximately \$16.39 million, or 16.39% of NAV, from February 19 to March 23, 2020. Severe Equity and Credit losses are partially offset by Treasury gains.

Table 9. COVID-19 Market Crash - Asset-Class Attribution

Asset Class	Historical Stress P&L	P&L as % NAV
Equity	-\$16.24m	-16.24%
Credit	-\$2.32m	-2.32%
Commodity	-\$0.18m	-0.18%
FX	-\$0.02m	-0.02%
Rates	+\$2.38m	+2.38%
Portfolio	-\$16.39m	-16.39%

Table 10. COVID-19 Market Crash - Position-Level Attribution

Instrument	Class	Exposure	Return	Stress P&L	% NAV
SPY	Equity	\$25.0m	-33.72%	-\$8.43m	-8.43%
QQQ	Equity	\$15.0m	-27.92%	-\$4.19m	-4.19%
VGK	Equity	\$10.0m	-36.26%	-\$3.63m	-3.63%
LQD	Credit	\$10.0m	-12.30%	-\$1.23m	-1.23%
HYG	Credit	\$5.0m	-21.90%	-\$1.09m	-1.09%
GLD	Commodity	\$5.0m	-3.62%	-\$0.18m	-0.18%
EURUSD	FX	\$3.0m	-0.94%	-\$0.03m	-0.03%
USDJPY	FX	\$2.0m	+0.49%	+\$0.01m	+0.01%
IEF	Rates	\$15.0m	+6.38%	+\$0.96m	+0.96%
TLT	Rates	\$10.0m	+14.23%	+\$1.42m	+1.42%

SPY generates the largest individual loss at -\$8.43 million. Although VGK has the largest percentage decline among the Equity positions, SPY's larger exposure drives a greater dollar contribution. The principal diversification benefit comes from Rates: IEF and TLT contribute a combined +\$2.38 million.

Large Equity and Credit Losses + Treasury Gains -> Partial Portfolio Loss Mitigation

4.3 2022 Inflation / Rates Sell-Off

The 2022 Inflation / Rates Sell-Off is the most severe historical scenario, producing a \$23.09 million loss, or 23.09% of NAV, from January 3 to October 14, 2022. Losses are broad across Equity, Rates, Credit, and Commodity.

Table 11. 2022 Inflation / Rates Sell-Off - Asset-Class Attribution

Asset Class	Historical Stress P&L	P&L as % NAV
Equity	-\$14.37m	-14.37%
Rates	-\$5.49m	-5.49%
Credit	-\$2.91m	-2.91%
Commodity	-\$0.46m	-0.46%
FX	+\$0.13m	+0.13%
Portfolio	-\$23.09m	-23.09%

Table 12. 2022 Inflation / Rates Sell-Off - Position-Level Attribution

Instrument	Class	Exposure	Return	Stress P&L	% NAV
SPY	Equity	\$25.0m	-24.27%	-\$6.07m	-6.07%
QQQ	Equity	\$15.0m	-34.77%	-\$5.21m	-5.21%
VGK	Equity	\$10.0m	-30.86%	-\$3.09m	-3.09%
TLT	Rates	\$10.0m	-30.59%	-\$3.06m	-3.06%
IEF	Rates	\$15.0m	-16.21%	-\$2.43m	-2.43%
LQD	Credit	\$10.0m	-21.74%	-\$2.17m	-2.17%
HYG	Credit	\$5.0m	-14.65%	-\$0.73m	-0.73%
GLD	Commodity	\$5.0m	-9.12%	-\$0.46m	-0.46%
EURUSD	FX	\$3.0m	-14.26%	-\$0.43m	-0.43%
USDJPY	FX	\$2.0m	+27.92%	+\$0.56m	+0.56%

Nine of ten positions generate negative P&L. Equity contributes -14.37% of NAV, Rates -5.49%, Credit -2.91%, and Commodity -0.46%. FX provides only a small +0.13% offset.

Equity Losses + Rates Losses + Credit Losses + Commodity Losses > FX Offset

Equity losses less in 2022 than during COVID-19 (-14.37% versus -16.24%), yet total portfolio loss is much worse (-23.09% versus -16.39%). The key difference is the Rates contribution, which swings from +2.38% during COVID-19 to -5.49% in 2022 - a 7.87% of NAV reversal.

4.4 2023 Regional Banking Stress

The 2023 Regional Banking Stress produces a positive portfolio result of approximately \$2.45 million, or 2.45% of NAV, from March 8 to March 24, 2023. Gains in Rates, Commodity, Credit, and selected Equity exposure more than offset limited losses elsewhere.

Table 13. 2023 Regional Banking Stress - Asset-Class Attribution

Asset Class	Historical Stress P&L	P&L as % NAV
FX	-\$0.01m	-0.01%
Credit	+\$0.37m	+0.37%
Equity	+\$0.38m	+0.38%

Asset Class	Historical Stress P&L	P&L as % NAV
Commodity	+\$0.45m	+0.45%
Rates	+\$1.27m	+1.27%
Portfolio	+\$2.45m	+2.45%

Table 14. 2023 Regional Banking Stress - Position-Level Attribution

Instrument	Class	Exposure	Return	Stress P&L	% NAV
VGK	Equity	\$10.0m	-2.00%	-\$0.20m	-0.20%
SPY	Equity	\$25.0m	-0.42%	-\$0.10m	-0.10%
USDJPY	FX	\$2.0m	-4.75%	-\$0.09m	-0.09%
HYG	Credit	\$5.0m	-0.33%	-\$0.02m	-0.02%
EURUSD	FX	\$3.0m	+2.69%	+\$0.08m	+0.08%
LQD	Credit	\$10.0m	+3.85%	+\$0.39m	+0.39%
GLD	Commodity	\$5.0m	+8.97%	+\$0.45m	+0.45%
TLT	Rates	\$10.0m	+4.94%	+\$0.49m	+0.49%
QQQ	Equity	\$15.0m	+4.55%	+\$0.68m	+0.68%
IEF	Rates	\$15.0m	+5.17%	+\$0.78m	+0.78%

Rates provide the largest positive contribution at +1.27% of NAV. GLD adds +0.45%; QQQ's gain is sufficient to offset losses in SPY and VGK, leaving Equity positive overall.

Treasury Gains + Gold Gains + Selected Equity and Credit Gains > Limited Losses Elsewhere

4.5 Historical Stress Findings

Historical outcomes vary materially across regimes. Equity remains the largest structural source of downside, but total severity depends critically on whether other exposures offset or reinforce Equity losses.

Table 15. Historical Rates Contribution Across Scenarios

Historical Scenario	Rates P&L as % NAV	Portfolio Role
COVID-19 Market Crash	+2.38%	Partial loss mitigation
2022 Inflation / Rates Sell-Off	-5.49%	Loss amplification
2023 Regional Banking Stress	+1.27%	Positive portfolio support

- 1. Equity concentration is the primary structural source of downside.**
- 2. Rates diversification is highly regime-dependent.**
- 3. Cross-asset loss alignment is a key determinant of portfolio stress severity.**

The position-level analysis also shows that the instrument experiencing the largest percentage decline does not necessarily generate the largest dollar loss; exposure size matters jointly with market-move severity.

5. Cross-Framework Risk Analysis

Historical and hypothetical stress provide complementary perspectives. Comparing them helps distinguish vulnerabilities that persist across forms of stress from those that depend on a particular scenario design or historical regime.

5.1 Historical vs. Hypothetical Scenario Comparison

Figure 4. Historical vs. Hypothetical Portfolio Stress Results

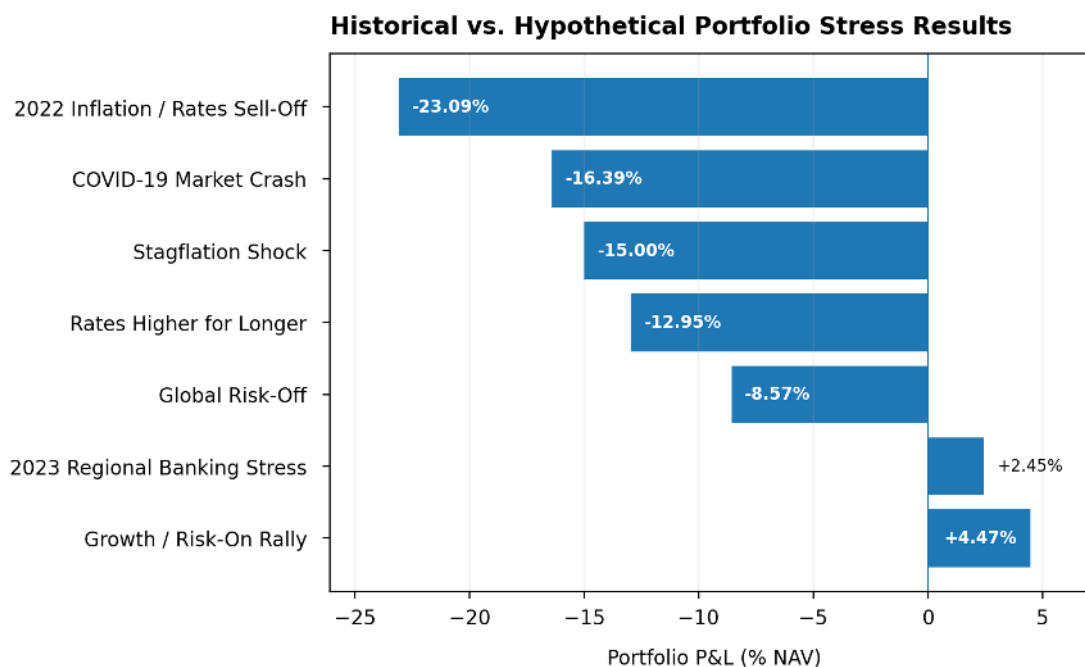
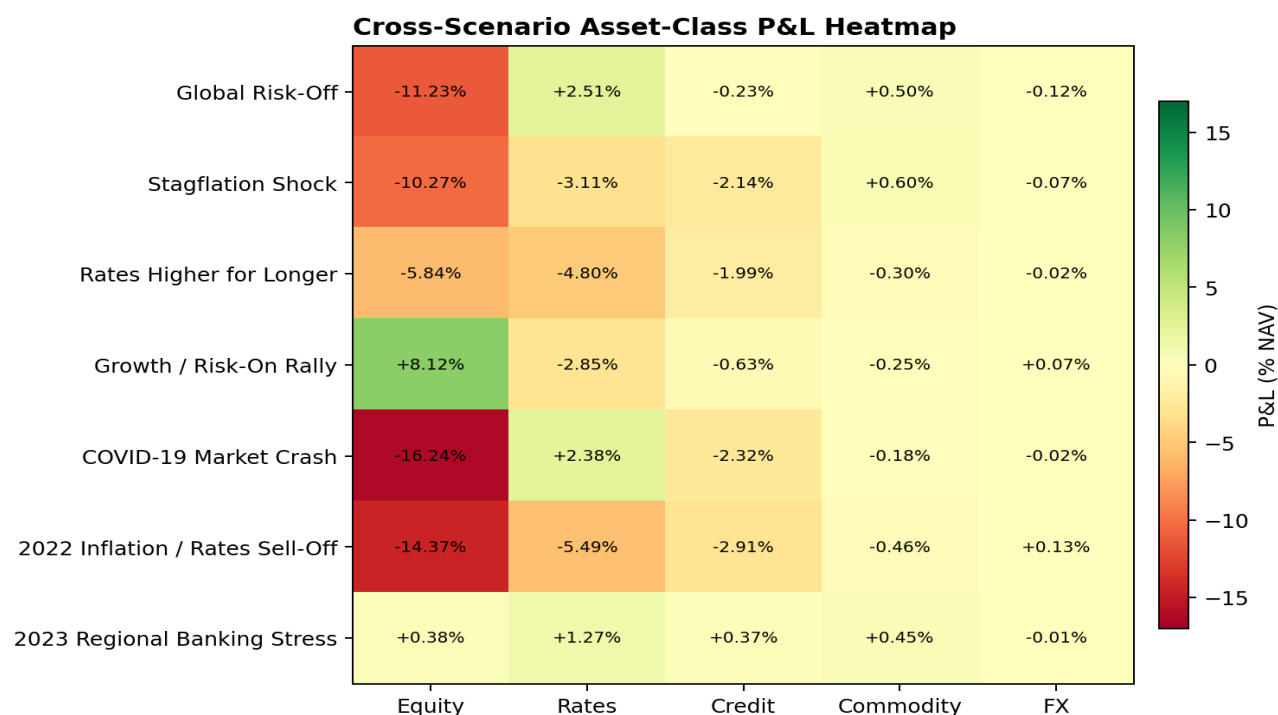


Table 16. Cross-Framework Scenario Comparison

Framework	Scenario	Portfolio P&L	P&L as % NAV
Historical	2022 Inflation / Rates Sell-Off	-\$23.09m	-23.09%
Historical	COVID-19 Market Crash	-\$16.39m	-16.39%
Hypothetical	Stagflation Shock	-\$15.00m	-15.00%
Hypothetical	Rates Higher for Longer	-\$12.95m	-12.95%
Hypothetical	Global Risk-Off	-\$8.57m	-8.57%
Historical	2023 Regional Banking Stress	+\$2.45m	+2.45%
Hypothetical	Growth / Risk-On Rally	+\$4.47m	+4.47%

The two most severe portfolio losses arise from different frameworks: 2022 Inflation / Rates Sell-Off is the worst overall at -23.09% of NAV, while Stagflation Shock is the most severe hypothetical scenario at -15.00%. This does not imply that historical scenarios are inherently more severe; the methodologies and shock specifications differ.

Figure 5. Cross-Scenario Asset-Class P&L Heatmap



The heatmap highlights cross-asset interaction. Rates contribute positively during COVID-19 and Global Risk-Off, but negatively during 2022, Stagflation, and Rates Higher for Longer. Total portfolio severity therefore cannot be inferred from Equity performance alone.

Conventional Risk-Off: Equity Weakness + Defensive Rates -> Partial Diversification

Inflationary Stress: Equity Weakness + Adverse Rates -> Diversification Breakdown

5.2 Integrated Portfolio Risk Assessment

Taken together, the analysis identifies three core characteristics: material Equity concentration, regime-dependent fixed-income diversification, and vulnerability to cross-asset loss alignment. These findings are visible under both historical replay and hypothetical stress.

Gold provides a useful but limited diversification source in several scenarios, while FX generally has a comparatively small total effect. Credit can remain contained or reinforce broader losses depending on the regime.

The principal implication is that portfolio resilience depends not only on diversification across asset-class labels, but on diversification across underlying economic risk drivers.

6. Model Limitations and Considerations

The hypothetical framework relies on deterministic scenario shocks and simplified risk-factor sensitivities. Primary sensitivities are generally 1.0, with selected additional exposures specified manually; these assumptions are not statistically estimated.

Fixed-income price effects are estimated using first-order modified-duration and spread-duration approximations. Convexity and other higher-order effects are not incorporated.

The analysis assumes a static portfolio. Current exposures remain constant, with no rebalancing, hedging, position changes, or management response during stress.

Historical results are sensitive to scenario-window selection. Different start and end dates around the same episode can materially change instrument returns and portfolio P&L.

Historical stress captures combined realized instrument repricing but does not identify the individual factors responsible for each movement. Hypothetical stress provides factor attribution but depends more heavily on mappings, sensitivities, and scenario assumptions.

The scenario set is selective rather than exhaustive. Liquidity effects, transaction costs, volatility-dependent behavior, nonlinear optionality, basis risk, and other second-order effects are not modeled explicitly.

Stress losses are conditional estimates under defined assumptions - not forecasts, probabilities, or estimates of maximum possible loss.

7. Conclusion

This report developed a multi-asset stress-testing framework to evaluate a \$100 million portfolio across both hypothetical and historical market environments. The combination of risk-factor-based hypothetical stress and historical replay provides complementary perspectives on portfolio vulnerability.

The results identify Equity concentration as the portfolio's largest structural source of downside. However, Equity weakness alone does not determine overall stress severity. The most damaging outcomes occur when Equity losses coincide with adverse performance in Rates, Credit, and other exposures, reducing the diversification benefits expected from a multi-asset portfolio. This pattern is particularly evident in the 2022 Inflation / Rates Sell-Off and the hypothetical Stagflation Shock.

Historical replay captures realized cross-asset behavior without requiring those relationships to be specified in advance, while hypothetical stress allows individual risk factors and economic assumptions to be examined explicitly. Used together, the frameworks provide a more complete assessment of portfolio vulnerability than either approach alone.

The central risk-management conclusion is that portfolio diversification should be evaluated not only by the number of instruments or asset classes held, but by the underlying economic risk drivers that those exposures share. Stress testing makes these relationships visible by identifying the conditions under which diversification remains effective and the conditions under which adverse cross-asset alignment can materially amplify portfolio losses.